

SURROS - 3 Month Execution Milestone Plan (Django Version)

Total Duration: 12 Weeks (3 Months)

Phases:

- 1) Foundation (Week 1–3)
- 2) Core System Build (Week 4–8)
- 3) Intelligence + Polish (Week 9–12)

TEAM ROLES:

- 1) Team Lead – Risk Engine + ML + Architecture
- 2) Data Engineer – Data + Preprocessing
- 3) Backend Dev – Django + API
- 4) Frontend Dev – UI + Charts
- 5) Research/Documentation Lead (optional)

PHASE 1 – FOUNDATION

Week 1: Research & Information Gathering

- Read 5 papers on urban risk & seasonal trends
- Finalize risk definition and pillars
- Collect datasets (traffic, accident, weather, pollution)
- Setup Django project & GitHub repo

Week 2: Data + Maths Implementation

- Clean dataset
- Apply Min-Max normalization
- Implement weighted risk formula
- Implement mean, variance, correlation, trend slope

- Create Django models (Zone, RiskData, Prediction)
- Basic dashboard UI setup

Week 3: OOP Structuring

- Refactor into classes (DataLoader, RiskCalculator, SeasonalAnalyzer, RiskPredictor)
- Create API endpoints (/zones, /risk/, /seasonal/)
- Integrate Chart.js for risk trend graph

PHASE 2 – CORE SYSTEM BUILD

Week 4: Seasonal Analysis Engine

- Monthly mean, variance, peak detection
- Seasonal API endpoint
- Highlight peak month in UI

Week 5: Prediction Model

- Implement Linear Regression
- Train-test split
- RMSE & MAE evaluation
- Display predicted next month risk

Week 6: Resource Recommendation

- Rule-based allocation logic
- Resource suggestion API
- Resource panel in dashboard

Week 7: Admin Panel

- Data upload functionality
- Model retrain trigger

- Admin dashboard UI

Week 8: Integration & Debugging

- Full system testing
- Fix bugs
- Data validation & edge case handling

PHASE 3 – INTELLIGENCE + POLISH

Week 9: Advanced Insights

- Correlation heatmap
- Multicollinearity check
- Insights page in dashboard

Week 10: Performance Optimization

- API response optimization
- Query performance tuning
- Code refactoring

Week 11: Documentation

- Architecture diagram
- Maths explanation
- Workflow diagram
- Dataset description
- Limitations & future scope

Week 12: Final Polish

- UI improvement
- Demo rehearsal

- Test scenarios
- Final presentation slides

Priority Focus Order:

- 1) Risk formula logic
- 2) Seasonal analysis
- 3) Clean OOP structure
- 4) Working dashboard
- 5) Prediction model

Execution Rule:

- Weekly Sunday review meeting
- Each member demo working code
- Working code > verbal progress